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Facial emotion recognition using Handcrafted features and CNN

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Abstract

Emotion recognition technology is a form of face detection and recognition that takes an aid of facial expressions and biophysical signs and symptoms like pulse rate and activations in the brain to actually know about an individual state. Image-based facial expression identification is a challenging topic, especially when it comes to assessing human emotion or mood in certain situations, such as while enjoying or watching a series, or movie, getting engrossed in video games, while shopping, and even on the battlefield. Emotions are of utmost concern due to an immediate increase in a number of healthcare concerns such as depression, cancers, paralysis, trauma etc. This research proposes an approach to emotion detection using feature extraction and convolution neural networks. Feature extraction techniques used are HOG and SIFT coupled with the convolutional neural network. Datasets considered for the training and testing of the proposed approach are CKplus and JAFFE . Experimental results show that HOG with CNN surpasses the state-of-the-art models.

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Keywords: Emotion recognition; Feature Extraction; HOG;SIFT

1. Introduction

When interacting and associating with others, emotions are expressed. Learning how to interpret them might be a difficult task, therefore technology is intended to be used as help. Amongst the numerous facial recognition techniques or technologies which have evolved majorly and increased over time is facial emotion recognition (FER). Currently, FER software is utilized to inspect and process human emotions in various extended applications like BCA (Brain Computer Interface).

Emotion Recognition (ER) has a lot of real-world applications. In the case of online video games, ER can be used to access the emotion or mood of the player, based on the emotion analyzed mode, levels, and the environment for the game can bechanged [1]. FER can be used in virtual training of the soldiers providing an analysis report of how the emotions change as a new challenge comes up on the battlefield. Recruiters use the FER for detecting if the candidate is genuine, honest, and keen to be in the applied position, and create a report for human recruiters to review, by assessing facial expressions. In digital marketing customers' emotions can be compared before and after entering the shopusing artificial intelligence cameras to evaluate how satisfied or elated the customers are with the service. Suppose if the score is poor, the application (hardware or software) might provide personal advice on how to enhance service quality.

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ER can detect and recognize various emotion by analyzing facial images. Lower and burrowed brows, as well as araised chin and concentrated gaze, can be used to identify rage, whereas elevated corners of the mouth into a smile can be used to determine joy. Figure 1 lists emotions and corresponding facial expressions that can be used to determine the emotions. FER majorly works on factors like placement of the eyebrows and mouth, structure of eyes, and distinct changes of the features related to face. With even minute changes in these mentioned parameters the classification of emotion changes.

Fig 1. Facial expressions for different emotions

Majority of existing strategies [2] categorize seven quite the usual or normal (prototypical) expressions that have been proven to be common across diversities and humanity: normal, pleased, surprised, frightened, furious, unhappy,and dislike. This paper proposes feature extraction with convolutional neural net as the FER methodology using the well-known datasets i.e. CK+ and Jaffe. The datasets involves major emotions such as frustration, wrath, happiness, furiousness, and other emotions.

The research and experimentation done is laid out as: The existing FER approaches and previously tried methods are briefly reviewed in Section 2. Section 3 showcase a quick look through the dataset used and the methodologies used for our research. The proposed FER technique's implementation is in Section 4. Experimentations and their studies are reflected in Section 5. The report finishes with a discussion of future research topics in Section 6.

2. Literature Review

Yu et al.(2018) suggested that to create effective and robust face expression analysis techniques, a substantial amount of labeled training data is necessary. However, with the rapid advancement of Internet services, availability, and Web technology, things are easier and possible to get hands-on an enormous number of photos with their respective information with minimal human work. Sun et al.(2017) came across a novel reduction way for dimension and an innovative classifier. The most common purpose of dimensionality reduction is to reduce within-class distances. Because FER systems are in an uncertain environment, benchmarked algorithms will eventually have to operate on databases from real encountered situations. Obtaining practical databases is quite crucial for benchmarked algorithms.

In Khan et al.(2016) face expression detection using local descriptors have proven to be somewhat a higher resistant to position changes, alignment issues, and effects of illuminations. A somewhat similar approach working on the positional parameters was implemented by Zhang et al.(2015) which stated that emotions can be detected

using an unrestrained automated facial feature or key element detection technique coupled with regression intensified based estimation for face active units (AUs) & clustered emotions. Kahoua et al.(2013) trained a convolutional neural net to have a look at the emotions from a static frame point of view with the help of 2 data followed by a per frame aggregation technique that employed the challenge training data. Muhamad et al.(2021) introduced a product face detection method based on deep learning that used real world datasets to achieve cutting-edge identification results. It is built and put into use a face recognition system using Faster R Convolutional Neural Network and Principal Component Analysis (PCA) (Faster R CNN).

In Minaee et al.(2021) a strategy of deep net based on an ANN which aims on finding the critical regions of the human face (highlighting the important features of the face) and outperformed earlier models on numerous datasets. In another research, a visualization-based method for the classifier's output to locate relevant and important facial areas for recognizing different emotions was used. Khaireddin et al.(2021) adopted VGGNet architecture, fine-tuned the hyperparameters with rigor, and experimented with various optimization strategies. Komla et al.(2020) suggested using Viola-algorithm John's to more correctly identify emotions. Real-time photos are taken, and characteristics are then retrieved from the facial photographs using this algorithm to track emotions in real-time. Sreevidya et al.(2022) developed an elderly-focused multi-modal system that incorporates information from audio and video modalities.

In Fard et al.(2022) the network was guided by an Ad-Corre Loss which showcase the intact feature and their corresponding vectors with strong correlatedness for inter-class samples. The decreased correlation for the in-between-class samples are also checked for correlations. There exist three components of Ad-Corre. Vulpe-Grigoraşi et al.(2021) proposed a strategy for improving the accuracy of face emotion identification was implemented by tweaking the hyperparameters of a convolutional neural network. The network's ideal hyperparameters were obtained by creating and training models using the Random Search method on a search space specified by discrete hyperparameter values. A CNN model for image categorization focusing on principal component analysis initialization was suggested by Ren et al.(2016).Rather than employing PCA to reduce down the dimensionality of the initial input, then PCA is made to work in effectiveness with the CNN training procedure. Suchinitialization values can lessen the influence of gradient diffusion problems caused by incorrect starting settings by containing image-altering information.

To focus on the face appearance detection problem, MF-MLP Classifier is presented in Sureddy et al.(2021).To illustrate the efficiency of the proposed classification model, extensive experiments were done on both restricted (CK+, JAFFE) and uncontrolled datasets (FER2013).A hybrid NN structure using Sparse Autoencoder and Convolutional Neural Network is developed (SCNN) by Yan et al.(2021) and to recognize human facial emotions, DNN or CNN are utilized in Gaddam et al.(2022). Using ResNet50, they have suggested a network for classifying human face expressions from static photos. In Mohammed et al.(2021) suggested a PCA approach for extracted features, and KNN was employed for classifying purposes.

3. Material and Methods

Despite a number of research going on in the field of ER, this field is yet to be discovered a better way to recognition of facial expressions and emotions. CNN, deep CNN,RCNN etc models are implemented and the recognizing accuracy has been quite promising. Deep learning has made the real-time emotion analysis an easier and more efficient task. Sarker et al.(2021) state that deep nets improve the accuracy of a model's performance. They enable a model to take in a collection of inputs and produce an output. The Deep Net enables a model to generalize on its own and then store those generalizations in a hidden layer known as the black box. This study is outlined majorly on feature extraction and training the model using CNN for CKplus and Jaffe datasets.

3.1 Dataset description

The CKplus dataset is available on Kaggle, and it contains 981 grayscale images with a resolution of 48x48 pixels. Images are organized into emotional folders. The dataset includes images of subjects within the age limit of 18 to 50 years, with a diversities of genders and ethnicities. Every image have been given labels from one of 7

variety of expression classes: rage, contempt, disgust, fear, happiness, sadness, and surprise.

The Jaffe dataset contains 213 labelled images of major expressions related to human objectified to natural conditions was collected from 10 unique individuals(female) from Japan. Each participant was requested to enact seven emotional expressions (six basic and one neutral), and the pictures were assessed by 60 validators, who assigned overall semantic values to each of the individual's facial expressions.

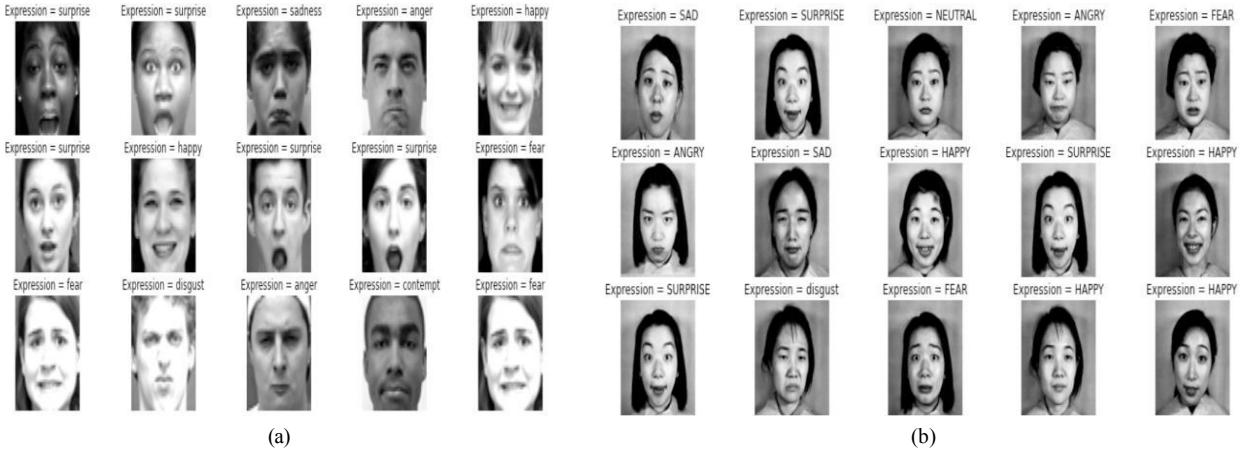


Fig 2. Data Visualization of (a)CKplus dataset (b)Jaffe dataset

3.2 HOG

The feature descriptor HOG is a technique that extracts characteristics from picture data. The HOG descriptor is concerned with the structure or form of an object. We just determine whether or not a pixel is an edge in the case of edge features. HOG can also be used to provide the edge direction. This is accomplished by extracting the edge gradient and orientation (or magnitude and direction). These orientations are determined in what are known as 'localized' portions. This implies the entire image is separated into several parts, each with its own gradients and orientation.

Before applying hog-features, the image must first be resized and converted to grayscale. After that, the image's gradient is determined, which is a combination of the image's magnitude and angle. After that, the gradient matrix is separated into cells of varying sizes to form blocks. After each block is produced, histogram computations are conducted, and normalization is the last procedure to be completed.

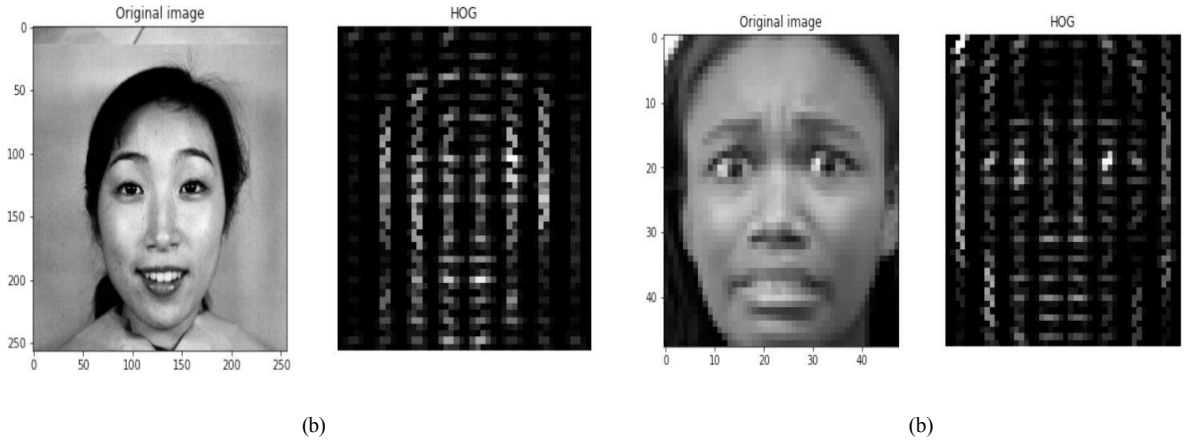


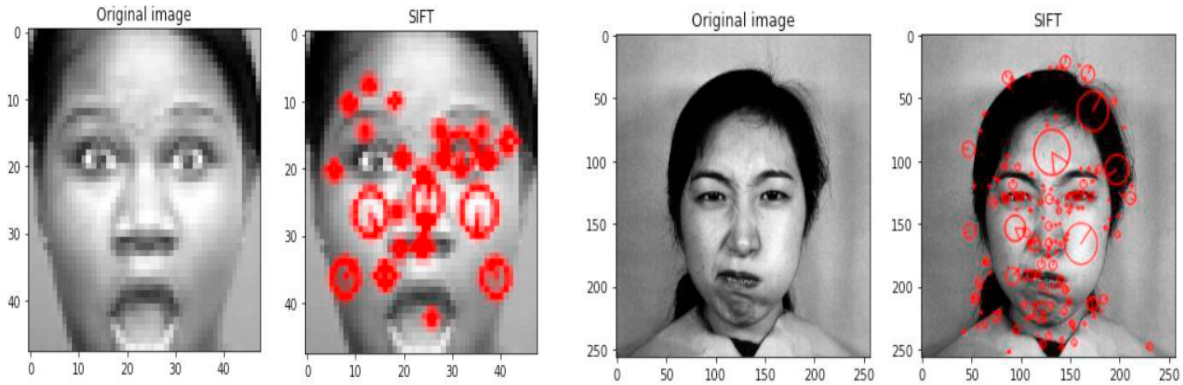
Fig 3. HOG feature representation for an image from (a) Jaffe dataset (b) CKplus dataset

3.3 SIFT

SIFT aids in the identification of the image's local features, often referred as 'key points.' Picture matching, object detection, scene detection, and other computer vision applications can all benefit from these scale and rotation

invariant key points. We use the key points obtained by SIFT as features for the image during model training. SIFT characteristics have the advantage of being unaffected by image size or orientation.

SIFT majorly works in four parts namely construction of scale space to make certain that features are adjusted - independent, to identify the characteristics, crucial point localization is used. that are of importance, orientation assignment is done to be sure that the key-features found are rotation invariants and at the end assigning uniquely maintained fingerprint to each of the key-feature.



(c) (b)

Fig 4. SIFT feature representation for an image from (a) CKplus dataset (b) Jaffe dataset

3.4 Proposed feature extraction-convolution network model

In this research emotion have been analyzed using the feature extraction as the initial step and then applying or feeding those extracted features into the convolution network formed using convolutional as well as dense layers. The feature extraction is performed using HOG and SIFT. CKplus and Jaffe dataset is used for the analysis purpose. The CKplus database contains 981 train images while Jaffe have 231 images. These images are then further disintegrated into the seven emotions in the dataset. Once the general pre-processing is complete the train image set is then passed through. The data from both the dataset is preprocessed and then individually feature extraction techniques are applied to bring the key features or the important features. These features involve essential information which main regards to the emotion detection process. Key-features extracted included information about the jawline structure, eyes are winked or mouth open etc. Mentioned parameters serve an important role in better analysis of emotions. The retrieved or detected features are then fed into a convolutional and dense layer neural network. Three consecutive convolutional layers are built into a sequential model, the output through which is then flattened. The flattened output serves as an input to the dense layers that are fully connected and have relu activations. The dense layers added into the sequential model have 128,64,16 and 7 neurons respectively. The output dense layer is structured having 7 neurons and a softmax classifier layer that provides the emotion dataset classifications.

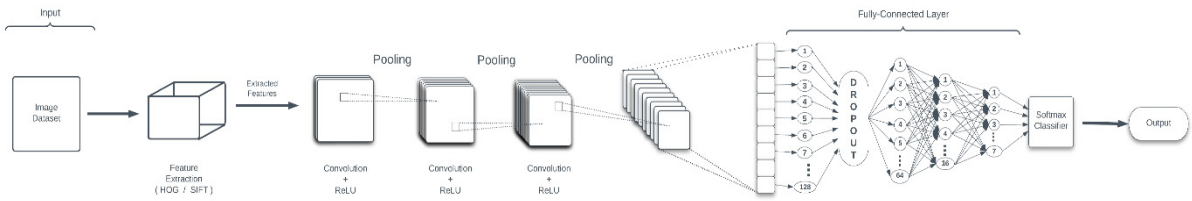


Fig 5. Proposed Architecture

4. Implementation

4.1 Data preprocessing

The train and test datasets both have enough data to recognize emotions. To have a clear understanding, the datasets are loaded and then the train and test data are separated into arrays. Using tensorflow.keras utils, the image labels of both train and test data are reshaped and translated to categorical values. The expand function is used to increase the size of the train and test images. The train and test images are then normalized separately. The HOG and SIFT feature extractions are then applied to the preprocessed images. These techniques are used to highlight important traits in order to improve detection.

4.2 Model training

The sequential model receives the preprocessed images. The convolutional network and the dense layer network are the two fundamental components of the model. Feeding the preprocessed photos into the three convolution net layers, each with its own maxpool layer. The convolution net's output is flattened before being transmitted to the dense layers, where dropout happens towards the end of the first dense layer. The softmax classifier produces categorized images as output. The training accuracies for both datasets have been quite promising, implying that the model has been properly trained.

4.3 Model testing

After the compilation of the created model, the model is tested for its computational performance for the testing dataset using the evaluate function. The model on testing reveals that it is capable of recognizing the emotions with an accuracy of 98.48% and 91.43% with HOG-CNN for CKplus and Jaffe respectively. While with SIFT-CNN the accuracy of 97.96% and 82.85% was recorded for CKplus and Jaffe dataset respectively.

5. Results and Discussion

Performance for the proposed approach and the comparison of our approach to the existing models will be seen in this section. Experimental procedures have already been outlined in the above section.

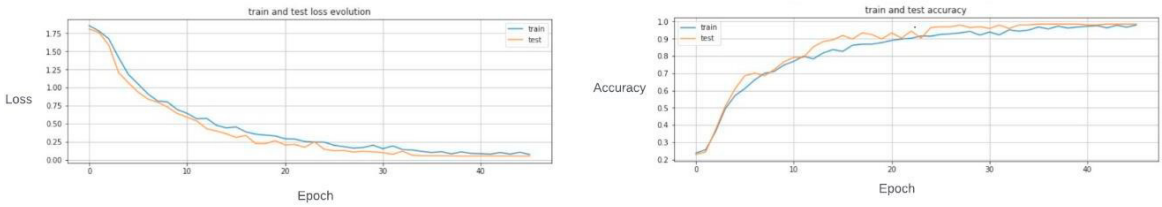


Fig 6. HOG-CNN for CKplus dataset

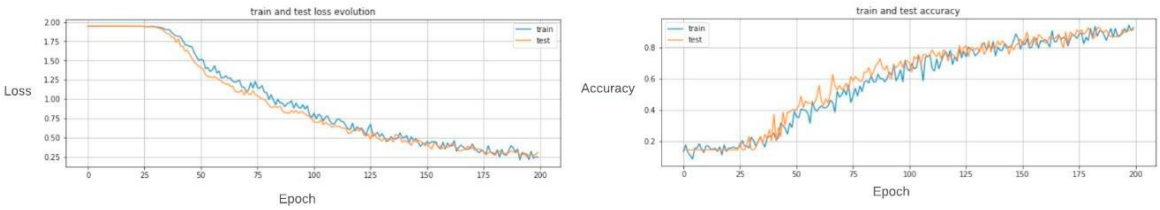


Fig 7. HOG-CNN for Jaffe dataset

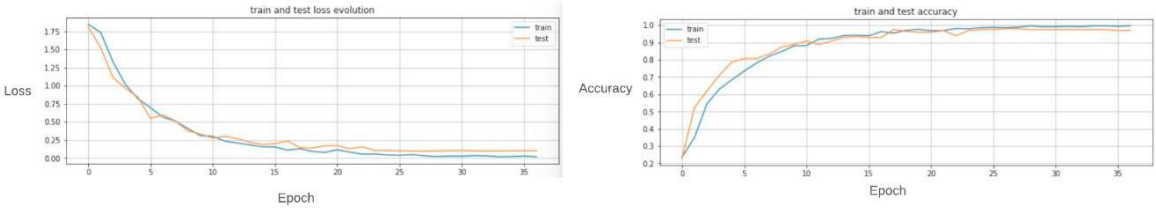


Fig 8. SIFT-CNN for CKplus dataset

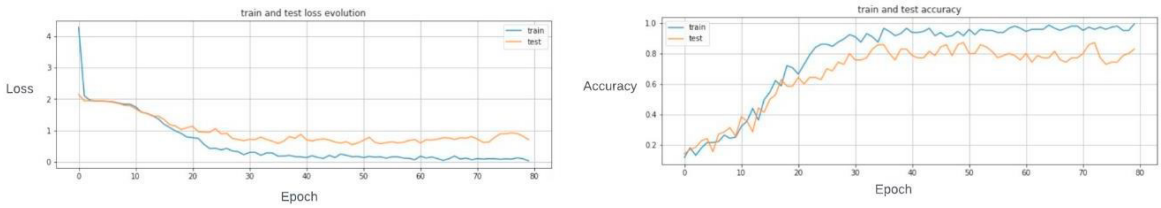


Fig 9. SIFT-CNN for Jaffe dataset

The experimental analysis for the proposed models for the datasets addressed is shown in Figures 6,7,8 and 9. The training and testing accuracies encountered on the proposed model are listed in Table 1. The confusion matrix for the HOG-CNN and SIFT-CNN models for both datasets is shown in Figure 10 and 11. Table 2 and 3 highlights a comparison between the proposed model approach and current models.

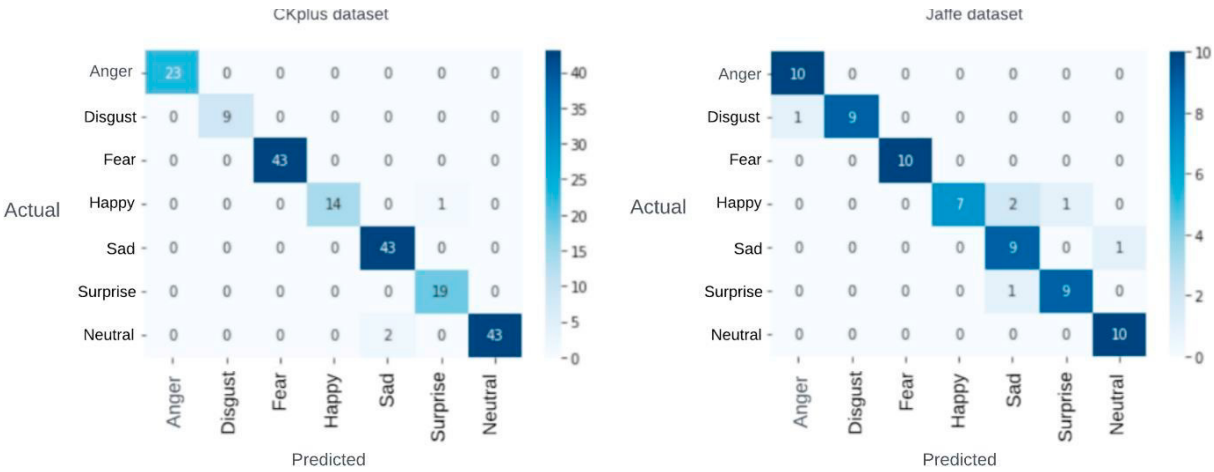


Fig 10. Confusion Matrix for HOG-CNN

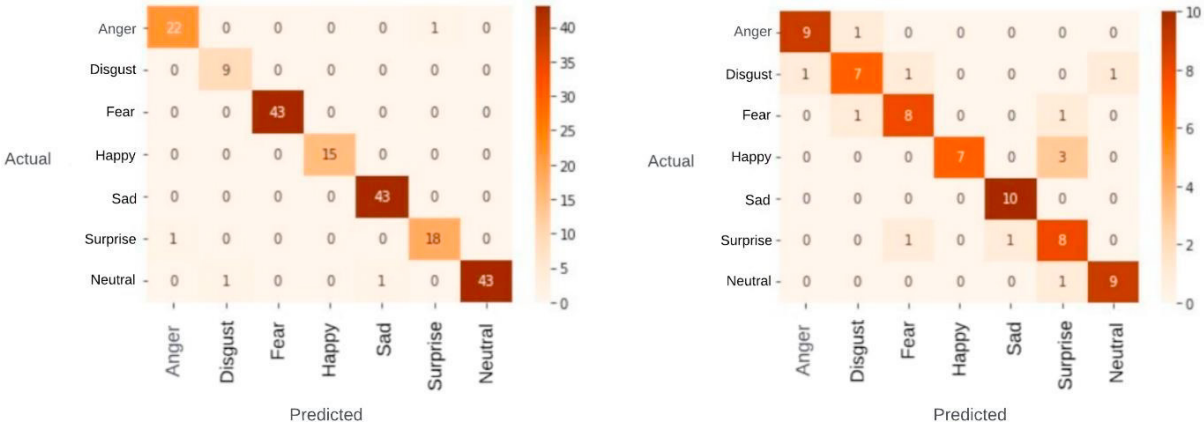


Fig 11. Confusion Matrix for SIFT-CNN

Table 1. Experimental results for HOG-CNN and SIFT-CNN

	Training Acc (CKplus)	Testing Acc (CKplus)	Training Acc (Jaffe)	Testing Acc (Jaffe)
HOG-CNN	97.96	98.48	93.01	91.43
SIFT-CNN	99.98	97.96	95	82.85

Table 2. Performance comparison with existing models on CKplus dataset

Approach	Accuracy
Nonlinear eval on SL + SSL puzzling [18]	98.23
DeepEmotion [8]	98.00
FN2EN [17]	96.80
HOG-CNN(Proposed model)	98.48
SIFT-CNN(Proposed model)	97.96

Table 3. Performance comparison with existing models on Jaffe dataset

Approach	Accuracy
HOG-SVM [19]	76.19
LBP-SVM [21]	76.23
Hierarchical Deep Neural Network [20]	91.27
HOG-CNN(Proposed model)	91.43
SIFT-CNN(Proposed model)	82.85

From table 2, it is evident that [18] is performing better than SIFT-CNN(proposed model) due to appropriate self-supervised co-training with a supervised activities. HOG-CNN(proposed model) has outperformed all the existing models. Handcrafted features extracted using HOG and SIFT have significantly helped in the FER process.

6. Conclusion

This research proposes a novel and effective framework for successful emotion recognition with the help of feature extraction and CNN. This study highlights that explicit key-feature extraction on the dataset can aid for an effective and efficient facial emotion analysis. Handcrafted feature extraction plays an important when you have a limited data collection and want to build the model without losing any significant or relevant information. The significant contribution/findings of the research carried out are: data feature extraction serves as a helpful procedure for accuracy improvement, reduces the risk of overfitting, speeds up the training process, improves data visualization and processing of data. Features are extracted using HOG and SIFT technique in this following research which are further used to train neural network and the architecture i.e. explicit feature extraction coupled with convolution net and additional deep neural layers have been the reason for this outperforming accuracy. The experimental results of the proposed model show that it is capable to recognize the emotions with an accuracy of 98.48% for HOG-CNN and 97.96% for SIFT-CNN for CKplus dataset. Also for the Jaffe dataset the accuracies encountered were 91.43% for HOG-CNN while 82.85% for SIFT-CNN. The proposed model can be validated on other challenging facial emotion datasets.

References

[1] Wikanningrum, A., Rachmadi, R. F., & Ogata, K. (2019, November). Improving lightweight convolutional

neural network for facial expression recognition via transfer learning. In *2019 International Conference on Computer Engineering, Network, and Intelligent Multimedia (CENIM)* (pp. 1-6). IEEE.

[2] Ng, H. W., Nguyen, V. D., Vonikakis, V., & Winkler, S. (2015, November). Deep learning for emotion recognition on small datasets using transfer learning. In *Proceedings of the 2015 ACM on international conference on multimodal interaction* (pp. 443-449).

[3] Yu, K., Wang, Z., Zhuo, L., Wang, J., Chi, Z., & Feng, D. (2013). Learning realistic facial expressions from web images. *Pattern Recognition*, *46*(8), 2144-2155.

[4] Sun, Y., & Wen, G. (2017). Cognitive facial expression recognition with constrained dimensionality reduction. *Neurocomputing*, *230*, 397-408.

[5] Khan, S. A., Hussain, A., & Usman, M. (2016). Facial expression recognition on real world face images using intelligent techniques: A survey. *Optik*, *127*(15), 6195-6203.

[6] Zhang, L., Mistry, K., Jiang, M., Neoh, S. C., & Hossain, M. A. (2015). Adaptive facial point detection and emotion recognition for a humanoid robot. *Computer Vision and Image Understanding*, *140*, 93-114.

[7] Kahou, S. E., Pal, C., Bouthillier, X., Froumenty, P., Gülçehre, Ç., Memisevic, R., ... & Wu, Z. (2013, December). Combining modality specific deep neural networks for emotion recognition in video. In *Proceedings of the 15th ACM on International conference on multimodal interaction* (pp. 543-550).

[8] Minaee, S., Minaei, M., & Abdolrashidi, A. (2021). Deep-emotion: Facial expression recognition using attentional convolutional network. *Sensors*, *21*(9), 3046.

[9] Khaireddin, Y., & Chen, Z. (2021). Facial emotion recognition: State of the art performance on FER2013. *arXiv preprint arXiv:2105.03588*.

[10] Fard, A. P., & Mahoor, M. H. (2022). Ad-Corre: Adaptive Correlation-Based Loss for Facial Expression Recognition in the Wild. *IEEE Access*, *10*, 26756-26768.

[11] Vulpe-Grigoraşi, A., & Grigore, O. (2021, March). Convolutional Neural Network Hyperparameters optimization for Facial Emotion Recognition. In *2021 12th International Symposium on Advanced Topics in Electrical Engineering (ATEE)* (pp. 1-5). IEEE.

[12] Ren, X. D., Guo, H. N., He, G. C., Xu, X., Di, C., & Li, S. H. (2016, June). Convolutional neural network based on principal component analysis initialization for image classification. In *2016 IEEE first international conference on data science in cyberspace (DSC)* (pp. 329-334). IEEE.

[13] Sureddy, S., & Jacob, J. (2021, May). Multi-features Based Multi-layer Perceptron for Facial Expression Recognition System. In *International Conference on Image Processing and Capsule Networks* (pp. 206-217). Springer, Cham.

[14]Yan, L., Sheng, M., Wang, C., Gao, R., & Yu, H. (2022). Hybrid neural networks based facial expression recognition for smart city. *Multimedia Tools and Applications*, *81*(1), 319-342.

[15] Gaddam, D. K. R., Ansari, M. D., Vuppala, S., Gunjan, V. K., & Sati, M. M. (2022). Human Facial Emotion Detection Using Deep Learning. In *ICDSMLA 2020* (pp. 1417-1427). Springer, Singapore.

[16] Mohammed, S. B., & Abdulazeez, A. M. (2021). Deep Convolution Neural Network for Facial Expression Recognition. *PalArch's Journal of Archaeology of Egypt/Egyptology*, *18*(4), 3578-3586.

[17] Ding, H., Zhou, S. K., & Chellappa, R. (2017, May). Facenet2expnet: Regularizing a deep face recognition net for expression recognition. In *2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017)* (pp. 118-126). IEEE.

[18] Pourmirzaei, M., Montazer, G. A., & Esmaili, F. (2021). Using Self-Supervised Auxiliary Tasks to Improve Fine-Grained Facial Representation. *arXiv preprint arXiv:2105.06421*.

[19] Eng, S. K., Ali, H., Cheah, A. Y., & Chong, Y. F. (2019, November). Facial expression recognition in JAFFE and KDEF Datasets using histogram of oriented gradients and support vector machine. In *IOP Conference series: materials science and engineering* (Vol. 705, No. 1, p. 012031). IOP Publishing.

[20] Kim, J. H., Kim, B. G., Roy, P. P., & Jeong, D. M. (2019). Efficient facial expression recognition algorithm based on hierarchical deep neural network structure. *IEEE access*, *7*, 41273-41285.

[21] Ravi, R., & Yadhukrishna, S. V. (2020, March). A face expression recognition using CNN & LBP. In *2020 Fourth International Conference on Computing Methodologies and Communication (ICCMC)* (pp. 684-689). IEEE.

[22] Sreevidya, P., Veni, S., & Ramana Murthy, O. V. (2022). Elder emotion classification through multimodal fusion of intermediate layers and cross-modal transfer learning. *Signal, Image and Video Processing*, *16*(5), 1281-1288.

[23] Komala, K., Jayadevappa, D., & Shivaprakash, G. (2020). Human Emotion Detection And Classification Using Convolution Neural Network. *European Journal of Molecular & Clinical Medicine*